Cold Boot Continuity as a Deployment Primitive

Verified Resumption Across Context Loss, Agent Substitution, and Runtime Migration

A successor may recover the work without pretending to remember what vanished or inherit authority it was never granted.

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Abstract

Long-running agentic work commonly appears continuous only while one conversation, process, provider, or context window remains available. When that surface ends, summaries and prompts attempt to reconstruct what mattered. They may preserve narrative direction while losing exact artifacts, unresolved alternatives, authority boundaries, custody, external state, intermediate evidence, and the reasons a transition was admitted or refused.

This paper proposes **Cold Boot Continuity** as a deployment primitive: the capacity for a newly instantiated runtime, beginning without the predecessor's conversational memory, to recover enough verified semantic, artifact, task, custody, authority, environment, and evidence state to resume the same governed work under an explicit claim ceiling.

Cold boot does not mean bit-identical internal cognition, deterministic continuation, or automatic authority transfer. A replacement model may reason differently, lack required capabilities, discover that the archive is stale, or legitimately refuse to continue. Continuity is instead preservation of identified obligations, exact heads, admissibility decisions, open alternatives, work custody, authority constraints, evidence debts, and revision lineage through a verified resumption protocol.

The paper distinguishes bootability, orientability, resumability, semantic continuity, authority validity, and evidenced continuity. It specifies a context capsule as a bounded projection from the Governed Meaning Base (GMB), not as the whole corpus or an executable grant. It also defines failure modes, adversarial tests, the universal realization funnel, and integration with project lineage, receipted reasoning, context governance, and work custody.

This is a conceptual architecture and verification program. The source names a JN reactor and a historical Constitutional Crystallization Framework (CCF);

this revision applies the author-confirmed terminology lineage `CCF -> CMB -> GMB` while preserving JN reactor identity as an unresolved system candidate. No named implementation is certified by this paper.

1. Impersistent Environments

An **impersistent environment** supports useful working state during an active interaction but does not reliably preserve the state required to continue the same governed process across context truncation, restart, handoff, migration, or provider change.

Impersistence is not the same as statelessness. A system may preserve chat history, files, embeddings, or server-side memory while still failing continuity because the retained material is incomplete, unauthoritative, unverified, or unavailable to the successor. Conversely, a system can lose the transcript yet resume safely if the governing state was externalized and verified.

Common failures include:

- context-window truncation removes qualifications and unresolved alternatives;
- summaries preserve conclusions while discarding their evidence or claim ceilings;
- intermediate artifacts disappear or cannot be distinguished from accepted heads;
- external services change after the snapshot;
- credentials, capabilities, or legal authority expire or are revoked;
- a successor lacks the predecessor's tools, dependencies, model capabilities, or environment;
- a handoff names the intended next step but not who holds custody or how results become visible;
- a saved archive contains stale instructions, prompt injection, secrets, or malicious artifacts.

Continuity must therefore be treated as a property of a whole resumption relation, not the presence of a summary file.

2. Definition and Non-Identity Claims

Let S_0 be the predecessor's last admitted state, P a cold-boot package or context capsule, R_1 a successor runtime, X_1 the external world observed at resumption, and G the governing resumption protocol. Define:

$$\mathrm{Resume}(S_0, P, R_1, X_1, G) \in \{\,\mathrm{RESUMED}, \mathrm{BOUNDED}, \mathrm{HOLD}, \mathrm{REFUSE}\}.$$

`RESUMED` means the successor has verified the required state and can continue within reacquired authority.

`BOUNDED` permits only a narrower surface. `HOLD` identifies missing evidence, capability, freshness, or authority. `REFUSE` records an incompatibility or unsafe condition.

Cold Boot Continuity does not assert:

- that the successor has the predecessor's private memory or subjective continuity;
- that different agents will produce the same reasoning path or output;
- that archive contents are true because they are archived;
- that an old authorization remains valid in a new runtime, time, jurisdiction, or effect class;
- that all tacit knowledge can be serialized;
- that a package can recreate remote services, people, hardware, markets, or institutions;
- that reconstructing a state proves the predecessor acted correctly.

The continuity claim is typed: **same governed work relation**, not same mind, process, or behavior.

3. Six Continuity Levels

Continuity claims should identify their level.

Level	Claim
Bootable	The package can be opened and its declared bootstrap procedure can execute in the target environment.
Orientable	The successor can identify exact heads, task, terminology, dependencies, open issues, and claim boundaries.
Resumable	The successor can perform a valid next transition or typed refusal without relying on the missing conversation.
Semantically continuous	The successor preserves relevant meanings, distinctions, alternatives, debts, and revision lineage rather than only filenames or conclusions.
Authority-valid	The successor independently establishes current authority, capability, jurisdiction, scope, and time for the requested work.
Evidenced	Tests and observations support all claimed lower levels for an exact package, runtime, environment, and external-state profile.

A package can be bootable but not orientable, orientable but not resumable, resumable but not authority-valid, or authority-valid for read-only inspection but not consequential effect. These distinctions prevent "it loaded" from becoming "continuity was achieved."

4. The Continuity State Model

A useful cold-boot package distinguishes at least the following state classes.

4.1 Exact Artifact State

Source heads, candidate artifacts, manifests, schemas, scripts, dependencies, configuration, and evidence files carry exact paths or logical names, hashes, representation formats, and predecessor relations. Hashes bind bytes under stated assumptions; they do not establish meaning or safety.

4.2 Semantic State

Concepts, definitions, invariants, claims, counterclaims, standing, relations, terminology rulings, conflicts, open questions, and revision history reside in a governed meaning surface. The current corpus architecture names this durable state the **Governed Meaning Base (GMB)**. Historical CCF and CMB names remain in source custody but are not three current systems.

4.3 Task and Workfront State

The active objective, completed transitions, queued work, stopping conditions, priorities, blockers, decision gates, and expected deliverables form the workfront. Completion claims must point to evidence rather than merely restate intent.

4.4 Custody State

The package identifies who or what currently holds operational responsibility for each unresolved transition, which leases or deadlines exist, what result-visibility route is required, and how custody can transfer. Custody is not ownership, permission, or authorship.

4.5 Authority and Capability State

Prior authority records are evidence of past standing, not automatically live grants. The successor must determine which permissions survive, which require renewed consent, which secrets or tokens are unavailable, and which operations remain forbidden.

4.6 Environment and External State

Tool versions, operating system, model/provider, enabled capabilities, network, services, repositories, databases, hardware, clock assumptions, and remote heads

can change independently of the package. Resumption requires reconciliation rather than pretending the snapshot is the world.

4.7 Evidence and Remainder

Verification results, observations, receipts, failed attempts, known blind spots, untested pathways, and unresolved counterevidence travel with their claim ceilings. A clean receipt chain does not prove complete state capture or successful external effect.

4.8 Ephemeral and Private State

Some conversational associations, tacit judgment, private chain-of-thought, secrets, personal data, or transient runtime state should not or cannot be preserved. The package declares those omissions. Cold boot recovers the state required for governed continuation, not every vanished internal event.

5. The Context Capsule

A **crystalline context capsule** is a bounded, purpose-specific projection assembled for a successor. It is not the whole GMB, the whole repository, the whole transcript, or an authority token.

```
ContextCapsule
  capsule_identity
  source_state_commitments
  target_task_and_audience
  selected_concepts_and_definitions
  exact_artifact_heads
  active_invariants_and_constraints
  open_alternatives_and_conflicts
  workfront_and_custody
  authority_history_and_required_reacquisition
  evidence_and_counterevidence
  environment_requirements
  known_omissions_and_external_dependencies
  expiry_and_revision_relation
  bootstrap_and_verification procedure
```

The capsule must preserve provenance from each projection element to durable state. Selection, ordering, compression, and omission are semantic transformations and require their own review. A capsule that says only "continue the prior work" is narratively suggestive but operationally weak.

6. Cold-Boot Protocol

A successor begins with no presumption that the package is complete, current, safe, or authoritative.

1. **Identify the package.** Verify manifest, schema, exact members, hashes, predecessor, declared target, and expiry.
2. **Inspect before execution.** Treat package text, code, and embedded instructions as untrusted source material until admitted under the current task and authority.
3. **Establish the runtime.** Record model or agent identity, capabilities, tools, environment, dependencies, and relevant limitations.
4. **Orient semantically.** Load the terminology rulings, definitions, invariants, current heads, open conflicts, claim ceilings, and task projection needed for this purpose.
5. **Reconstruct the workfront.** Identify the active objective, completed evidence-backed transitions, outstanding work, custody, stopping rules, and pending open editorial questions.
6. **Reacquire authority.** Determine which requested operations are currently permitted. Do not inherit publication, deployment, messaging, credential use, or consequential-effect authority from a historical capsule.
7. **Reconcile external state.** Compare remote heads, services, databases, schedules, locks, human decisions, and other mutable dependencies with their recorded state.
8. **Run verification.** Execute package, environment, semantic, and task-specific checks, including negative and tamper cases where proportionate.
9. **Classify readiness.** Return `RESUMED`, `BOUNDED`, `HOLD`, or `REFUSE` with explicit reasons and remainder.
10. **Make the next transition.** Record the successor's exact input state, decision, result, evidence, new debts, and visibility route.

An `orient` command can implement part of this protocol. Command success alone is not continuity; the command must expose the required state and be tested against stale, missing, conflicting, and adversarial packages.

7. Agent Substitution Without Determinism

Agent substitution is valuable because work need not depend on one provider, model, process, or conversation. But substitution has limits.

Different agents may vary in reasoning ability, tool access, safety policy, context capacity, latency, nondeterminism, and interpretation. The proper invariant is not identical output. It is preservation of the governed problem and obligation surface.

For a transition request q, successor agents R_a and R_b may produce different candidates y_a and y_b while both remaining admissible:

$$y_a \neq y_b, \qquad y_a, y_b \in \text{AdmissibleContinuations}(S_0, q).$$

Where exact replay is required, the package must specify a deterministic or bounded-equivalence relation, captured dependencies, seeds, tool outputs, clocks,

and environmental assumptions. Many research and writing tasks instead require semantic and constitutional continuity, not trajectory identity.

A less capable successor may legitimately return `HOLD` or

`REFUSE`. That is stronger continuity than confident fabrication.

8. Archive-Centered Computation, Properly Bounded

The source offers a productive inversion:

> In an archive-centered system, the session interprets and revises governed state rather than treating state as a disposable by-product of conversation.

The archive is not automatically operational ground truth. It may be incomplete, stale, corrupted, malicious, internally inconsistent, or contradicted by external state. GMB entries may have different standing. Candidate artifacts are not current heads. A receipt may bind a past decision without making it correct today.

The more precise architecture is **state-centered, evidence-bounded computation**. Durable state supplies the predecessor; orientation and verification establish which parts can bind the successor; external reconciliation supplies current conditions; authority evaluation determines permitted continuation; revision writes the new relation.

Conversational history can remain useful as nonauthoritative source evidence, rhetorical context, or a place to recover alternatives not yet crystallized. Structural continuity should not require the whole transcript, but neither should crystallization erase relevant uncertainty simply because it is inconvenient to encode.

9. Failure Modes

Cold-boot systems can fail through:

- **package incompleteness:** a runtime-required file, dependency, schema, or key is missing;
- **head ambiguity:** multiple candidates are presented without a current-head or conflict relation;
- **semantic compression loss:** qualifiers, counterevidence, alternatives, or standing disappear;
- **authority laundering:** past consent or a textual instruction is treated as a current grant;
- **custody gap:** no eligible actor owns the next operational responsibility;
- **result invisibility:** work completes in a runtime or location the requester cannot inspect;

- **environment drift:** tools, providers, libraries, hardware, or platform behavior differ materially;
- **external-state drift:** remote data, services, schedules, or human decisions changed after packaging;
- **orientation drift:** the bootstrap selects a different meaning, head, or task than the package intended;
- **candidate contamination:** exploratory or hostile material enters durable state without admission;
- **split-brain succession:** multiple successors act as the sole continuation from one predecessor;
- **secret or privacy leakage:** the archive preserves material that should not travel;
- **self-verification:** the package's own claims and verifier fail together while appearing consistent;
- **nontermination:** recursive dependency or legitimacy checks never reach a typed disposition;
- **archive growth:** each generation carries all predecessors until the package becomes impractical.

Each failure should produce a typed result and recovery obligation rather than an optimistic summary.

10. Evidence and Adversarial Tests

A cold-boot claim should be tested with an exact package and a genuinely fresh runtime that lacks the predecessor conversation.

Core tests include:

1. **Fresh-agent resumption:** a new agent identifies the task, heads, constraints, open issues, and next valid transition.
2. **Cross-provider substitution:** a materially different runtime preserves the obligation surface or reports a bounded incapability.
3. **Tamper test:** changed, missing, extra, relocated, duplicate, oversized, or case-colliding members are detected before unsafe execution.
4. **Stale-state test:** changed remote or external state forces reconciliation rather than silent reuse.
5. **Revoked-authority test:** a previously authorized operation is refused after revocation or expiry.
6. **Candidate-standing test:** speculative and AI-synthesized material cannot silently become author-approved or canonical.
7. **Semantic-loss test:** omitted qualifications or counterevidence are detected by dependency and source checks.
8. **Capability-mismatch test:** missing tools or model abilities produce a typed hold instead of simulated success.
9. **Split-brain test:** parallel successors receive distinct lineage and cannot both claim exclusive continuity.

10. **Result-visibility test:** the output and evidence arrive at an inspectable route named by the work order.
11. **Recovery test:** interruption during the first successor transition can be retried or rolled back without duplicating effect.
12. **Minimality test:** removing redundant history does not change the verified resumption result, while removing a load-bearing element does.

The evidence package should bind package identity, predecessor, runtime, environment, tests, failures, observations, verifier, and unresolved remainder. One successful demonstration does not establish general portability.

11. Continuity Profile

Cold Boot Continuity should not be reduced to one score. Report a profile:

$$\mathbf{C} = (B, O, R, S, A, E, V),$$

where B is bootability, O orientability, R resumability, S semantic preservation, A current authority validity, E environment and external-state reconciliation, and V verification strength.

Each component carries its own method, scope, evidence, and confidence. A deployment can set minimum floors based on consequence. High bootability cannot compensate for absent authority; strong hashes cannot compensate for stale external state; good orientation cannot compensate for an uninspectable result.

Useful operational measures include time to orientation, unsupported-claim rate, missed dependency rate, false-head selection, authority leakage, successful next-transition rate, repair time, package size, boot cost, cross-runtime divergence, and proportion of known obligations preserved.

12. Realization Funnel

Stage	Cold-boot continuity responsibility
Possibility	Explore persistence, archive, transcript, snapshot, capsule, replay, and migration designs, including unsafe and overbroad ones.
Specification	Define continuity levels, state classes, dispositions, identity, authority, custody, evidence, external-state, and privacy obligations.
CM1	Enumerate admissible packages and protocols that satisfy the bare specification.

Stage	Cold-boot continuity responsibility
CM2	Prefer minimal, inspectable, provider-portable, privacy-preserving, authority-safe, tamper-evident, repairable, and cost-bounded configurations.
Instance Configuration	Select exact schema, packager, verifier, runtime targets, context compiler, stores, authority process, tests, and visibility route.
Implementation Realization	Produce the exact package, bootstrap, verifier, target runtime, environment, and operational handoff.
Evidence	Execute fresh-runtime, tamper, stale-state, revocation, semantic-loss, split-brain, recovery, visibility, and minimality tests.
Revision	Update state, capsule schema, dependencies, authority, environment profiles, or tests while preserving predecessor and migration relations.

The paper specifies the primitive and candidate protocol. It does not certify the JN reactor or any capsule as an evidenced Instance Configuration.

13. Corpus Integration

Cold Boot Continuity binds several corpus structures without merging them:

- **GMB** stores durable, standing-aware concepts, claims, relations, conflicts, decisions, and revision lineage. A capsule is compiled from GMB for a task.
- **A Constitution for the Context Window** governs selection, omission, ordering, provider realization, and claim ceilings in the successor's invocation context.
- **Governed Context & Memory** distinguishes context configuration, durable memory, and explicit ignorance.
- **The Three Hash Rule** separates exact state, admissibility commitments, and meaning commitments in the package.
- **Reasoning by Way of Receipts** preserves decisions, refusals, debts, and intermediate evidence without converting receipts into truth.
- **Governed Semantic Project Lineage** binds exact project heads, builds, configurations, predecessor relations, rebuildability, and migrations.
- **A Formal Theory of Work Custody** governs responsibility, lease, handoff, visibility, and recovery across successor agents.

- **Semantic Legitimacy Before Context** prevents archived or retrieved material from acquiring binding standing merely by being present.
- **The Constitutional Observability Platform** supplies heterogeneous verification, external observation, evidence-health, and remainder reporting.
- **Hyperpedia** is a human- and agent-facing knowledge projection; **Crystal Network** is a distributed exchange surface. Neither is automatically the runtime archive or authority source for every cold boot.

The source's **JN reactor** remains an explicitly named system candidate requiring identity and implementation review. Its historical relation to CCF, current GMB architecture, and other reactor or archive systems is not automatically merged by this paper.

Verified Resumption Envelope

The completed P1 corpus gives Cold Boot Continuity one precise role: it is a **verified resumption envelope and protocol for reconstructing bounded work state across context loss, agent substitution, process restart, or runtime migration**. It does not recreate vanished experience, make successor agents identical, or transfer authority and capability by semantic resemblance.

- **Adjacent work:** *A Formal Theory of Work Custody*
  Supplies: Responsibility for exact work heads, transfer, return, and terminal disposition.
  Cold-boot boundary: A capsule can carry custody records; a successor accepts custody only through a valid transfer.

- **Adjacent work:** *Meaning Crystals*
  Supplies: Portable semantic packages with reconstruction obligations.
  Cold-boot boundary: A crystal may be capsule content; semantic portability is not full operational resumption.

- **Adjacent work:** *Sovereign Software*
  Supplies: Reconstructability, provider removal, migration, and effective exit across dependency closure.
  Cold-boot boundary: Cold boot is one continuity mechanism and test within a broader sovereignty claim.

- **Adjacent work:** *Reasoning by Way of Receipts*
  Supplies: Bounded records of decisions, alternatives, failures, and revisions.
  Cold-boot boundary: Receipts help reconstruct why work reached its state; they do not restore ambient authority or external reality.

- **Adjacent work:** *Continuation Primacy*
  Supplies: The analytic priority of what must persist for continuation.
  Cold-boot boundary: Cold boot turns selected continuation obligations into an artifact and protocol.

A resumption envelope should bind:

$$R = (artifact_state, semantic_state, workfront, custody, \\ authority_requirements, capabilities, environment, . \\ evidence, remainder, private_omissions, expiry)$$

The envelope records **requirements and prior grants**, but resumption must revalidate current authority, capabilities, external state, environment, time, and revocation. A prior authorization may be evidence that an act was once permitted; it is not necessarily a live grant for a successor, host, or later invocation.

Continuity claims should state their level: byte recovery, structural recovery, semantic orientation, task resumption, custody resumption, authorized operational resumption, or observed outcome continuity. Success at a lower level does not imply a higher one. Reproducible bootstrap establishes only the levels exercised by its bound tests and witnesses.

The universal GSM can type `PACKAGED_IN_CAPSULE`, `RESUMES_WORK_HEAD`, `RECONSTRUCTS_SEMANTIC_STATE`, `TRANSFERS_CUSTODY_BY`, `REQUIRES_REAUTHORIZATION`, `REBINDS_CAPABILITY`, `REVALIDATES_ENVIRONMENT`, `RECONCILES_EXTERNAL_STATE`, `OMITS_PRIVATE_STATE`, `EXPIRES_AT`, and `LEAVES_REMAINDER`. These relations prevent context continuity from laundering identity or permission.

This integration establishes four editorial commitments: adopt verified resumption envelope as the official corpus role; adopt the resumption-envelope schema; publish continuity as a levelled claim; and require fresh authority, capability, environment, and external-state validation before operational resumption.

Semantic Service and Capability Rebinding

The fixed-seed comparison with Semantic Debt, Semantic Computing Infrastructure, the Cognitive ABI, and Crystallized Governance extends operational resumption:

$$R^* = (capsule, SCI_roots, service_profiles, open_debt, \\ semantic_remainder, capability_contracts, \\ governance_artifacts, current_grants, environment, \\ evidence, reconciliation)$$

The capsule should bind the GMB and universal GSM roots, atlas or registry versions, SCI service profiles, open semantic-debt entries, and unresolved remainder required to reconstruct the work. Recovery of files without those liabilities can resume the wrong semantic state while appearing byte-complete.

Each callable operation must be rebound through the Cognitive ABI. The successor rechecks conformance, qualification, admission, current authority, permitted effects, and witnessability for the present host and invocation. A remembered provider name, successful import, or prior receipt is not current capability or permission.

Crystallized-governance artifacts may carry source correspondence, explicit rules, remainder, contest, amendment, and fork identity. They should be restored as governed inputs, not as self-executing authority. Current adoption, interpretation, jurisdiction, revocation, and effect conditions still require validation.

A high-level resumption proof therefore demonstrates a ladder: exact capsule verification; governed semantic-state reconstruction; open-debt and remainder recovery; service and capability rebinding; current governance and authority validation; external-state reconciliation; bounded execution; and observed effect evidence. Failure at a higher rung does not erase lower-level recovery.

The universal GSM can type `RESTORES_SCI_STATE_ROOT`, `RECOVERS_OPEN_SEMANTIC_DEBT`, `REBINDS_CAPABILITY_CONTRACT`, `REVALIDATES_GOVERNANCE_ARTIFACT`, `REFUSES_STALE_AUTHORITY`, `RECONCILES_EXTERNAL_DEPENDENCY`, and `PROVES_RESUMPTION_LEVEL`.

This randomized review establishes four editorial commitments: include semantic-service roots, debt, and remainder in operational capsules; require Cognitive ABI rebinding for every callable capability; treat restored governance crystals as inputs rather than live grants; and adopt the runged resumption proof.

A Compiled Continuity Capsule Plus Present-Tense Reacquisition

Fourteen retained downstream uses place cold boot across context-window governance, work custody, capability ABIs, the historical CMB-to-current-GMB lineage, split authority, operational and general epistemic orientation, project lineage, model-first migration, semantic infrastructure, handoff control, semantic debt, governed knowledge loops, and admissibility. They reveal a two-part system: a custodied capsule preserves what can travel, while a cold-boot protocol revalidates everything that must be true now.

$$\begin{aligned} C^* = (&predecessor_head, capsule_root, semantic_roots,\\ &workfront, custody_claims, authority_requirements,\\ &capability_requirements, environment_profile,\\ &external_state, privacy_omissions, debt, remainder,\\ &bootstrap, verification, successor_head, evidence,\\ &expiry, revision) \end{aligned}.$$

The capsule is a purpose-bounded projection from durable sources such as GMB, artifact stores, lineage records, and ledgers. It is not the durable meaning base, vanished cognition, a complete transcript, or live authority. Exact verification establishes only exact custody. Semantic reconstruction, task orientation, custody acceptance, capability rebinding, present authority, external-state reconciliation, bounded execution, and observed outcome form higher and non-substitutable proof rungs.

Adaptive boot is allowed only through typed repair and successor creation. Missing providers, schema changes, unavailable tools, revoked grants, moved resources, private omissions, and environmental drift may yield bounded degradation, refusal, or a revised capsule; they must not be silently normalized. Successor identity records continuity without asserting agent identity or deterministic behavior.

The universal GSM can type
`COMPILES_COLD_BOOT_CAPSULE`, `VERIFIES_EXACT_CAPSULE`,
`RECONSTRUCTS_SEMANTIC_ROOT`, `ACCEPTS_WORK_CUSTODY`, `REBINDS_PRESENT_CAPABILITY`,
`REACQUIRES_CURRENT_AUTHORITY`, `RECONCILES_EXTERNAL_STATE`,
`DEGRADES_WITH_TYPED_REMAINDER`, and
`CREATES_COLD_BOOT_SUCCESSOR`.

This review establishes four editorial commitments: formalize the capsule/protocol split; adopt the complete cold-boot record and runged proof; require present-tense authority, capability, environment, and external-state reacquisition; and make adaptive repair, degradation, refusal, expiry, and successor identity explicit.

A Capsule Is Not Continuity Until It Rebinds to the Present

Public orientation

Cold-boot continuity lets a newly instantiated agent reconstruct authoritative state, verify lineage, and resume bounded work without depending on conversational memory. A portable capsule carries claims about prior state. Continuity is established only when those claims are verified and the new environment lawfully reacquires present capabilities, authority, and result visibility.

Adversarial tests

1. **Stale-capsule attack.** The archive is internally valid but superseded by later decisions or revocations. Freshness and successor discovery precede resumption.
2. **Authority-replay attack.** A prior mandate is reused after scope, time, issuer, or jurisdiction changed. Authority must be reacquired or explicitly shown continuous.
3. **Capability-drift attack.** Named tools, models, APIs, paths, or permissions differ in the new environment. Rebinding requires compatibility and

consequence review.

4. **Secret-custody attack.** The capsule embeds credentials or relies on unavailable secrets. Sensitive material needs separate custody; absence must produce refusal, not substitution.
5. **Non-idempotent replay.** Resuming repeats an external effect already performed. Receipts, effect identity, idempotency, and reconciliation are mandatory.
6. **Archive-confusion attack.** Duplicate paths, traversal, case collisions, symlinks, or executable startup hooks alter the restored subject. Extraction must be hostile and layout-bound.
7. **Result-visibility gap.** Work completes in a runtime the user or next agent cannot inspect. Continuity includes return, evidence, and discoverable terminal state.
8. **Semantic-compression loss.** A compact capsule omits assumptions, disputes, failures, or open decisions. Compression must preserve typed remainder and retrieval routes.

Claim and evidence ceiling

The paper specifies a verified resumption envelope. It does not prove operational continuity until a cold process reconstructs the exact subject, verifies custody, reacquires present authority and capabilities, avoids duplicate effects, performs bounded work, and returns visible evidence.

Claims about portability, security, reliability, compression, cross-runtime compatibility, or deployment performance require exact archives, hostile extraction tests, multiple clean environments, and independent reproduction. Internal corpus fit is not a live test. No external research was added in this pass.

Reader-facing consequence

The public standard is that an agent can carry its research state between sessions without smuggling stale authority or trusting memory: the new session can prove what it received, what changed, what it may do now, and where its results went.

14. Limits and Counterevidence

The proposal is weakened or requires revision if:

- a small conventional checkpoint plus task file provides equivalent recovery at lower cost;
- agents cannot reliably interpret the semantic capsule across providers or model generations;
- required tacit knowledge consistently fails to externalize;
- packaging and verification cost exceeds the value of resumability;
- privacy and secret-management constraints make the archive unsafe;

- exact reconstruction encourages stale continuation rather than current reconsideration;
- authority cannot be safely reacquired without the original human or institution;
- external-state drift dominates archived state;
- deterministic replay is impossible for the target task and bounded semantic equivalence is not reliable;
- archive-centered workflows centralize control or prevent legitimate plural successors;
- repeated clean boots do not improve task success, repair time, or unsupported-claim rate over summary baselines.

A current external review is required across workflow checkpointing, event sourcing, durable execution, build reproducibility, provenance, software supply chain, distributed systems, capability security, agent memory, context compression, human-computer handoff, privacy, and disaster recovery. Internal corpus consistency does not validate comparative or implementation claims.

15. Conclusion

Cold Boot Continuity relocates long-running work from the lifespan of a conversation into a governed relation among durable state, a bounded context capsule, a successor runtime, current external conditions, reacquired authority, custody, and evidence.

The archive is necessary but not sovereign. The capsule is useful but incomplete. The successor is substitutable but not identical. A hash binds bytes but not truth. A handoff preserves history but not permission. A successful bootstrap proves less than a successful governed next transition.

The deployment primitive is therefore not "load this summary and continue." It is:

> Begin without presumed memory; verify the predecessor; recover the work's semantic and artifact state; reconcile the present world; reacquire authority; classify what can continue; and record the successor relation.

When that protocol works across fresh runtimes and adversarial conditions, agentic work can survive context windows, provider changes, interruption, and migration without pretending that vanished cognition persisted. Continuity becomes structural, but it remains bounded, testable, revisable, and honest about what could not be carried forward.

Realization Scope and Evidence Boundary

This successor makes the paper's stage boundary explicit. It narrows claims rather than manufacturing missing evidence: a stage named as context is not thereby claimed as implemented, validated, or complete.

- **Possibility Space.** The present paper does not claim a complete possibility space treatment. That stage remains successor work and cannot be inferred from the architecture described here.
- **Revision.** The present paper does not claim a complete revision treatment. That stage remains successor work and cannot be inferred from the architecture described here.
- **Publication form.** This work is retained as a standalone technical companion because its concrete migration method is useful independently; it remains subordinate to the architecture it explains.

Corpus-Wide Reference Positioning

This paper argues that long-running agentic work commonly appears continuous only while one conversation, process, provider, or context window remains available. Its nearest external comparison fields are storage, provenance, observation. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: Fixity and preservation can keep bytes stable while representation dependencies, executability, interpretation, and institutional custody decay. A complete provenance graph can faithfully record a false or unauthorized process; provenance is evidence about lineage, not automatic truth or legitimacy. Telemetry can be complete at the instrumented boundary while omitting the causal variable, real-world effect, or authority question that matters. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] Consultative Committee for Space Data Systems. *Reference Model for an Open Archival Information System (OAIS).* CCSDS 650.0-M-3, Issue 3, December 2024. https://ccsds.org/Pubs/650x0m3.pdf **Relationship:** Predecessor preservation model for information packages, representation information, and designated communities.

[**E2**] John Kunze et al. *The BagIt File Packaging Format (V1.0).* RFC 8493, 2018. https://www.rfc-editor.org/rfc/rfc8493 **Relationship:** Adjacent custody and transfer format for payload manifests, fixity, and complete package movement.

[**E3**] Juan Benet. "IPFS - Content Addressed, Versioned, P2P File System." 2014. arXiv:1407.3561. https://arxiv.org/abs/1407.3561 **Relationship:** Predecessor content-addressed network architecture; content identity is compared with, not treated as sufficient for, semantic jurisdiction.

[**E4**] Reproducible Builds Project. *Formal Definition of Reproducible Builds.* reproducible-builds.org source **Relationship:** Adjacent reproducibility practice; byte reproduction is compared with, but not equated to, semantic or constitutional continuity.

[**E5**] W3C. *PROV-O: The PROV Ontology.* W3C Recommendation, 30 April 2013. https://www.w3.org/TR/prov-o/ **Relationship:** Adjacent provenance standard; it models entities, activities, and agents, but does not by itself establish the paper's stronger authority or standing claims.

[**E6**] Santiago Torres-Arias et al. "in-toto: Providing Farm-to-Table Guarantees for Bits and Bytes." *28th USENIX Security Symposium*, 2019, pp. 1393-1410. usenix.org source **Relationship:** Predecessor supply-chain lineage system used for comparison; no influence claim is made unless the source head already documents one.

[**E7**] OpenSSF. *Supply-chain Levels for Software Artifacts (SLSA), Version 1.2.* 2026. https://slsa.dev/spec/v1.2/ **Relationship:** Adjacent software-supply-chain assurance standard, especially relevant to provenance and verification boundaries.

Ableman corpus

[**A1**] Adam Ableman. *Constitutional Storage*. Ableman Corpus current edition; work identity `PAPER.PUB.20788662`;
source SHA-256
`115725C7F7B72C3B720E4EF67F660F329B778D34C4D4232848C6082E849203F7`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Semantic Custody*. Ableman Corpus current edition; work identity `PAPER.PUB.20300952`;
source SHA-256
`E3887FE98E6369E39AC3FC2EAC63F2DF7C2AD18AD682B3A745C0CD0AD241D0F0`.
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[**A3**] Adam Ableman. *From Content-Addressed Storage to Governed Semantic Project Lineage*. Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-F1682AE03F0BA815`;
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[**A4**] Adam Ableman. *The Implementation Codex*. Ableman Corpus current edition; work identity
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Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Reasoning by Way of Receipts*. Ableman Corpus current edition; work identity
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Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *The Three Hash Rule*. Ableman Corpus current edition;

work identity PAPER.PUB.20788617;
source SHA-256
F6FA73D083B70711A8C817F7779751D1506FEBA0E356DB9805BB1570663731B5.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Editorial source notes retained from predecessor

Adam Ableman. *A Constitution for the Context Window.* Ableman Corpus current edition; conceptual claims retain the standing stated in the work.

Adam Ableman. *A Formal Theory of Work Custody.* Ableman Corpus current edition; conceptual claims retain the standing stated in the work.

Adam Ableman. *The Constitutional Observability Platform.* Ableman Corpus current edition; conceptual claims retain the standing stated in the work.

Adam Ableman. *From Content-Addressed Storage to Governed Semantic Project Lineage.* Ableman Corpus current edition; conceptual claims retain the standing stated in the work.

Adam Ableman. *Governed Context & Memory (GCM): Context Configuration as Constitutional Surface, Semantic Hashing as Enforcement, and Explicit Ignorance as First-Class Success in Generational AI.* Zenodo, 2026. https://doi.org/10.5281/zenodo.18633797.

Adam Ableman. *Governed Meaning Base (GMB).* Ableman Corpus current edition in the author-confirmed historical terminology CCF -> CMB -> GMB; conceptual claims retain the standing stated in the work.

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Adam Ableman. *Semantic Legitimacy Before Context.* Ableman Corpus current edition; conceptual claims retain the standing stated in the work.

Adam Ableman. *The Three Hash Rule.* Zenodo, 2026. https://doi.org/10.5281/zenodo.20788617.

External technical and scholarly research remains required for a future evidence-qualified revision.

The entries below are corpus-internal comparison and dependency sources. Their inclusion does not establish external validation of this paper's empirical, safety, novelty, or performance claims.

Adam Ableman *The Three Hash Rule.* Zenodo, 2026. doi:10.5281/zenodo.20788617

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Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus
License: CC-BY-ND-4.0

DOI: 10.5281/zenodo.23847490